

# **Pharmaceutical Calculations (Phar 2072)**

**(For Year II Pharmacy students)**

# **1. Some fundamentals of measurements and calculations**

# Objectives of the session

- At the end of the class you will be able to:
  - define balance sensitivity
  - define significant figures
  - describe rules of significant figures
  - define accuracy and precision in measurements
  - mention instruments for measurement of volume and weight
  - calculate percentage of errors in measurements

# Balance sensitivity

- ***Balance sensitivity*** may be defined as the minimum weight a balance can weigh accurately.
- All measuring balances do not show the same reading for weighing very small quantity of substances.
- A balance which has a capacity of measuring a very small quantity accurately is more sensitive than a balance which cannot measure very accurately.

## Significant figures

- ***Significant figures*** are digits that have practical meaning in measurements.
- It is also defined as one that is actually measured using the measuring device.
- Any number used in a measurement should contain only figures or digits that are considered reliable.

## Significant figures.....cont'd

- In a measurement, significant figures in a number consist of: *Figures (digits) definitely known + one approximated figure (digit) (usually the final digit).*
- One of the factors determining the degree of approximation to perfect measurement is the precision of the instrument used.
  - For example, It would be incorrect to claim that *7.76 milliliters had been* measured in a graduate calibrated in units of *1 milliliter*, or that *25.562 grams had been weighed* on a balance sensitive to 0.01 gram.

## Significant figures.....cont'd

### □ Rules of significant figures

- Nonzero digits are significant.
- Zero between any of the other digits in a number is a significant figure.
  - In reading the measurement 9.04 cm, the zero represents a measured quantity, just as 9 and 4, and is, therefore, a significant number.

## Significant figures.....cont'd

- Zero used only to show the location of decimal point is not significant.
  - In reading the measurement 0.46 cm, the zero does not represent a measured quantity, but merely locates the decimal point. It is not a significant figure.
- Final zeros after a decimal point are significant.
  - In reading the measurement 11.30 cm, the zero is an estimate and represents a measured quantity. It is therefore significant.

## Significant figures.....cont'd

- Zeros at the end of a whole number may or may not be significant.
  - Reporting measurements in scientific notation removes all doubt, since all numbers written in scientific notation are considered significant.
  - If 1600 is written as  $1.6 \times 10^3$  has two significant figures; if it is written as  $1.60 \times 10^3$ , has three significant figures; when it is written as  $1.600 \times 10^3$  it has four significant figures.

# *Accuracy and Precision*

- **Accuracy** is a measure of how closely the result of a measurement agrees with the standard accepted value.
  - It is the *trueness* or *exactness* of a measurement.
- **Precision** is the *uniformity* or *reproducibility* of a measurement.
  - If you take the same amount of substance, say 100 mg, and weigh it 100 times, you will not always end up with the same reading. Why not? Air pressure, humidity, temperature, dust, breezes, – all these things can affect the way the balance “reads”.

# ***1.2 Measurement of volume***

- **Common instruments for the pharmaceutical measurement of volume include:**
  - ✓ **cylindrical graduates**
  - ✓ **conical (cone-shaped) graduates**
  - ✓ **micropipettes (*for quantities of one ml or less*)**
  - ✓ **pipettes**
  - ✓ **burettes**
  - ✓ **calibrated vessels**
  - ✓ **medicinal droppers (*for small quantities*)**

# *Measurement of volume.....cont'd*

*Cylindrical graduates*



*Conical (cone-shaped) graduates*



# Measurement of volume.....cont'd

- **Micropipettes**



- **Medicinal droppers**



- **Burettes**



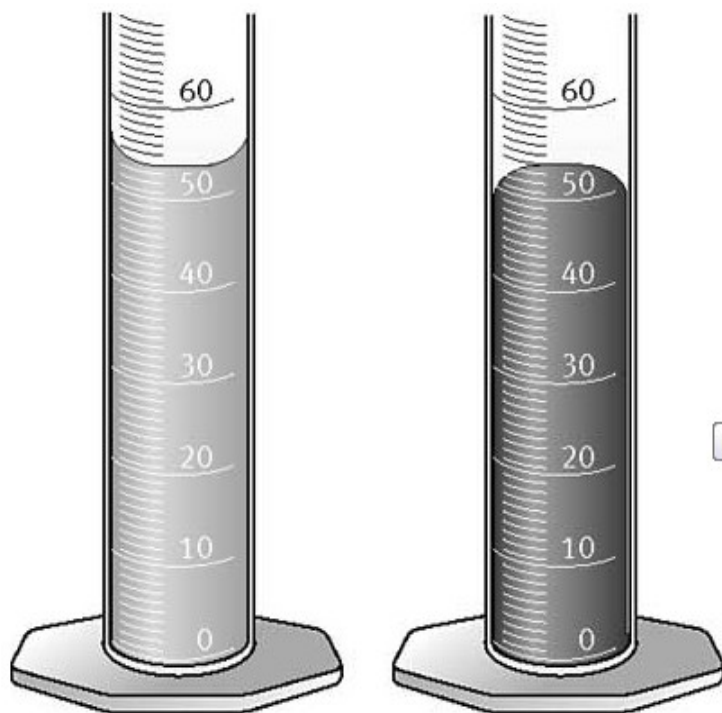
## Measurement of volume.....cont'd

- The selection of measuring instrument should be based on the level of accuracy required.
- ***As a general rule, it is best to select the graduate with a capacity equal to or just exceeding the volume to be measured.***
- But measurement of small volumes in large graduates tends to increase the size of the error.

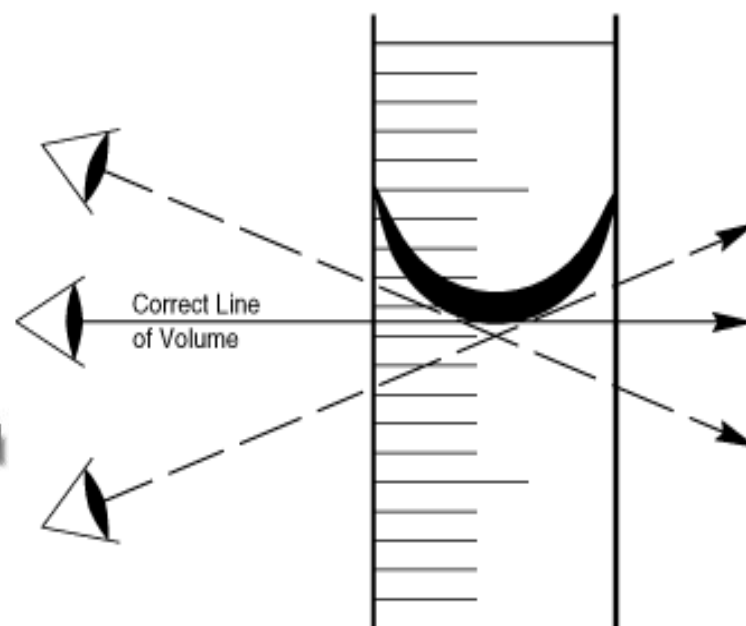
# *Measurement of volume...*



## *Measurement of volume.....cont'd*



meniscus



# Measurement of weight

- The type of balances used in pharmacy practices include digital balances (**electronic analytical balances**).
- Capable of weighing accurately 0.1 mg or less, equipped with convenient digital readout features.

--- Has greater accuracy



# Percentage of errors

- When a pharmacist measures a volume of liquid or weighs a material, two quantities become important:
  - ✓ The *apparent weight* or *volume measured*,  
*and*
  - ✓ The *possible excess* or *deficiency* in the actual quantity obtained.

# Percentage of errors.....cont'd

- **Percentage of error** may be defined as the maximum potential error multiplied by 100 and divided by the quantity desired.
- The calculation may be formulated as follows:

$$\text{percentage of error} = \frac{\text{Error} \times 100}{\text{desired weight or desired volume}}$$

$$\text{Error} = \text{Actual weight or volume} \\ - \text{desired weight or volume}$$

# Examples

1. A pharmacist weighed 475 milligrams of a substance on a balance of dubious accuracy. When checked on a balance of high accuracy, the weight was found to be 445 milligrams. Calculate the percentage of error in the first weighing. **Ans: 6.32%**
2. A graduate weighs 35.825 grams. When 10 milliliters of water are measured in it, the weight of the graduate and water is 45.835 grams. Calculate the weight of the water and express any deviation from 10 grams as percentage of error. **Ans: 0.1%**

## **1.3 Aliquot method of weighing and measuring**

# Aliquot method of weighing and measuring volume

- In measuring weight or volume, when percentage error is 1% or less, the measurement is termed as **highly accurate**.
- Measurements resulting in percentage errors between 1% and 5% are **moderately accurate**,
- But methods of **low accuracy** produce percentage errors greater than 5%.
- In pharmacy, like other practices, the maximum allowable error for measurements is 5%.

## Aliquot method of weighing and measuring volume (cont'd)

- When one attempts to weigh a substance whose weight is closer to the **sensitivity requirement** of a balance, the percentage error occurring may be greater than 5%.
- ***Sensitivity requirement*** is defined as the load that will cause a change of one division on the index plate of the balance.

# Aliquot method of weighing and measuring volume (cont'd)

- In order to avoid error of more than 5 % in a measurement, we usually measure the **smallest measurable quantity**.
- What do we do if the required measurement is below the smallest measurable quantity?
  - The aliquot method of weighing and measuring volume enables us to weigh or measure volume of a substance below its smallest measurable quantity.

# Aliquot method of weighing

- The aliquot method of weighing is a method by which small quantities of a substance may be weighed within the desired degree of accuracy by weighing a larger than needed portion of the substance, diluting it with an inert material, and then weighing a portion( aliquot) of the mixture calculated to contain the desired amount of the needed substance.

## Aliquot method of weighing (cont'd)

- Lactose is the most commonly used diluent in solid dosage form pharmaceuticals such as powders, tablets and capsules. Some of the reasons included are its extremely low incidence of side effects, ready availability, and low cost.

# Aliquot method of weighing (cont'd)

- The steps in aliquot method of weighing includes:

- ✓ Determine the smallest weighable quantity for the instrument in use;

$$\text{Smallest Weighable quantity (SWQ)} = \frac{\text{Sensitivity requirement} \times 100\%}{\text{Acceptable error}(\%)}$$

- ✓ Weigh the smallest weighable quantity or any multiple of the desired quantity of the drug greater than SWQ.

## Aliquot method of weighing (cont'd)

- ✓ dilute with determined quantity of an inert substance to obtain a dilution or mixture.
- the following formula can be used to calculate the weight of dilution/mixture and hence the weight of diluent needed.

$$\frac{\text{Weight of drug in dilution}}{\text{Weight of dilution}} = \frac{\text{weight of drug in aliquot}}{\text{weight of aliquot}}$$

## Aliquot method of weighing (cont'd)

- ✓ Finally, weigh an aliquot (part, portion) of the dilution, which contained the desired quantity of the substance.

# Examples

1. prescription calls for 50 milligrams of chlorpheniramine maleate. Using a prescription balance with a sensitivity requirement of 6 milligrams, explain how you would obtain the required amount of chlorpheniramine maleate with an error not greater than 5%. Use lactose as a diluent.

# Examples...

2. A prescription balance has a sensitivity requirement of 4 milligrams. Explain how you would weigh 5 milligrams of hydromorphone hydrochloride with an error not greater than 5%. Use lactose as the diluent.

## Examples...

3. A formula calls for 0.6 milliliter of a coloring solution. Using a 10-milliliter graduate calibrated from 2 to 10 milliliters in 1-milliliter units, how could you obtain the desired quantity of the coloring solution by the aliquot method? Use water as the diluent.

# Examples...

4. Using a 10-milliliter graduate calibrated in 1-milliliter units, explain how you would measure 1.25 milliliters of a dye solution by the aliquot method. Use water as the diluent.

## **1.4 Density, specific gravity, specific volume**

# Objectives of the class

- At the end of the class you will be able to:
  - define density, specific gravity, and specific volume
  - do calculations related to density, specific gravity, and specific volume

# Density

- Density is defined as mass per unit volume of a substance at a fixed temperature and pressure.

$$\text{Density (g/mL)} = \frac{\text{mass (g)}}{\text{Volume (mL)}}$$

- Example, If 100 ml of Lugol's solution weighs 120 g, its density is **1.2 g/mL**

# Specific gravity (sp gr)

- Specific gravity is the ratio of density of a substance to density of a standard, the value of both substances being determined at the same temperature or at another specified temperature.
- A pharmacist uses sp gr to convert the weight of an ingredient or preparation to volume or *vice versa*.
- In practice, liquids are the object of conversion.

## Specific gravity (sp gr) (cont'd)

- Water is used as the standard for the *sp gr* of *liquids and solids*; the most useful standard for gases is hydrogen.
- For practical purposes, *sp gr* may be defined as the ratio of the weight of a substance to the weight of equal volume of water.

$$\text{sp. gr of a substance} = \frac{\text{Weight of a substance}}{\text{weight of equal volume of water}}$$

# Specific volume (sp. volume)

- **Specific volume (sp. volume)** is the ratio of the volume of a substance to the volume of equal weight of another substance taken as standard, the volume of both substances being determined at the same temperature.
- The standard for liquids and solids is water.

# Specific volume (sp. volume) (cont'd)

$$\text{Sp. volume of a substance} = \frac{\text{Volume of a substance}}{\text{Volume of equal weight of water}}$$

- *Note:* The specific gravity and specific volume are reciprocal of each other.

# Examples

1. If a liter of cough syrup weighs 1285 g, what is its specific gravity? Ans: 1.285
2. What is the weight, in kilograms, 5 liters of sulfuric acid with specific gravity of 1.84?  
Ans: 9.2 kg
3. If 73.42 g of a liquid measures 81.5 mL, what is its specific volume? Ans: 1.11